

Quantum Search Algorithm

→ Begins at all $|0\rangle$ state

All entries $\rightarrow |x\rangle$

Search Register $\rightarrow$ superposition of all $\rightarrow \sum_x a_x |x\rangle$
solution)

Step-1 $\rightarrow$ Quantum Circuit for searching

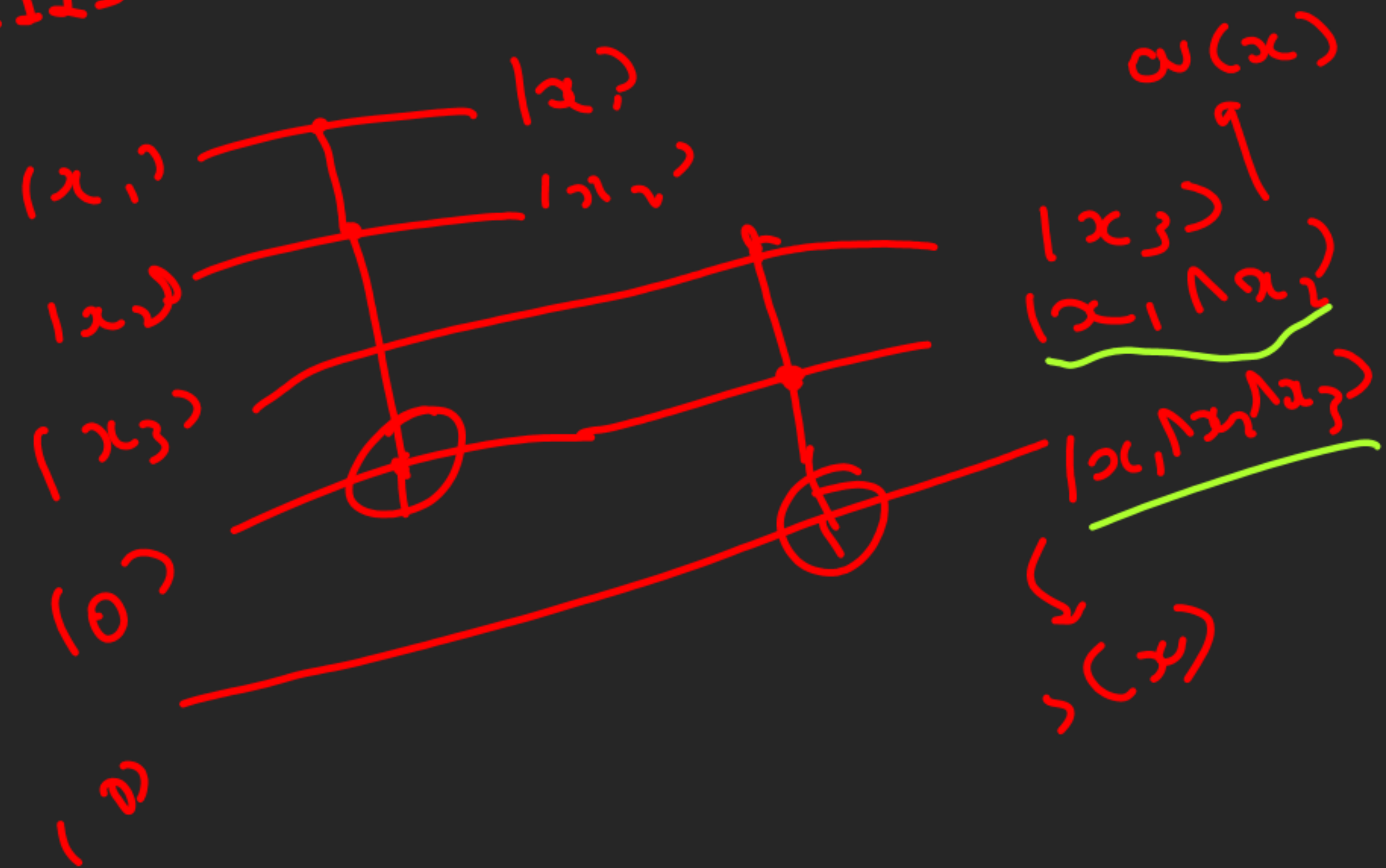
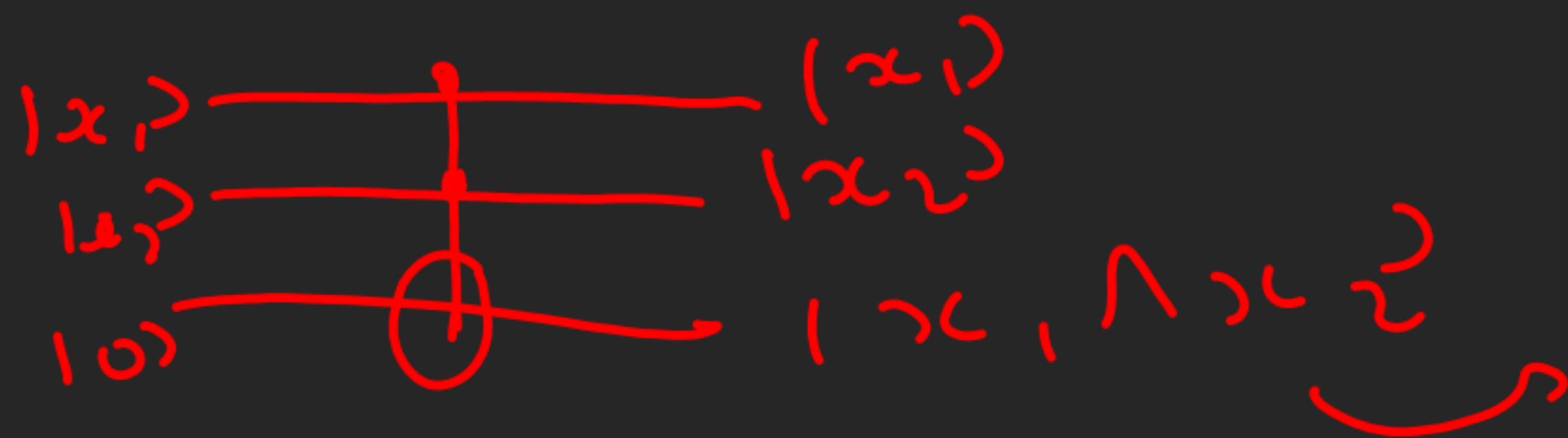
Next $\rightarrow$ check if it is a solution

Search Oracle / Black Box

$C_S |x\rangle |0\rangle \rightarrow |x\rangle |S(x)\rangle \rightarrow$ There's a complication

Say, $S(x) = x_1 \wedge x_2 \wedge x_3$ $|x\rangle = |x_1, x_2, x_3\rangle$

↳ solution $\rightarrow |111\rangle$



Quantum OR Gate

